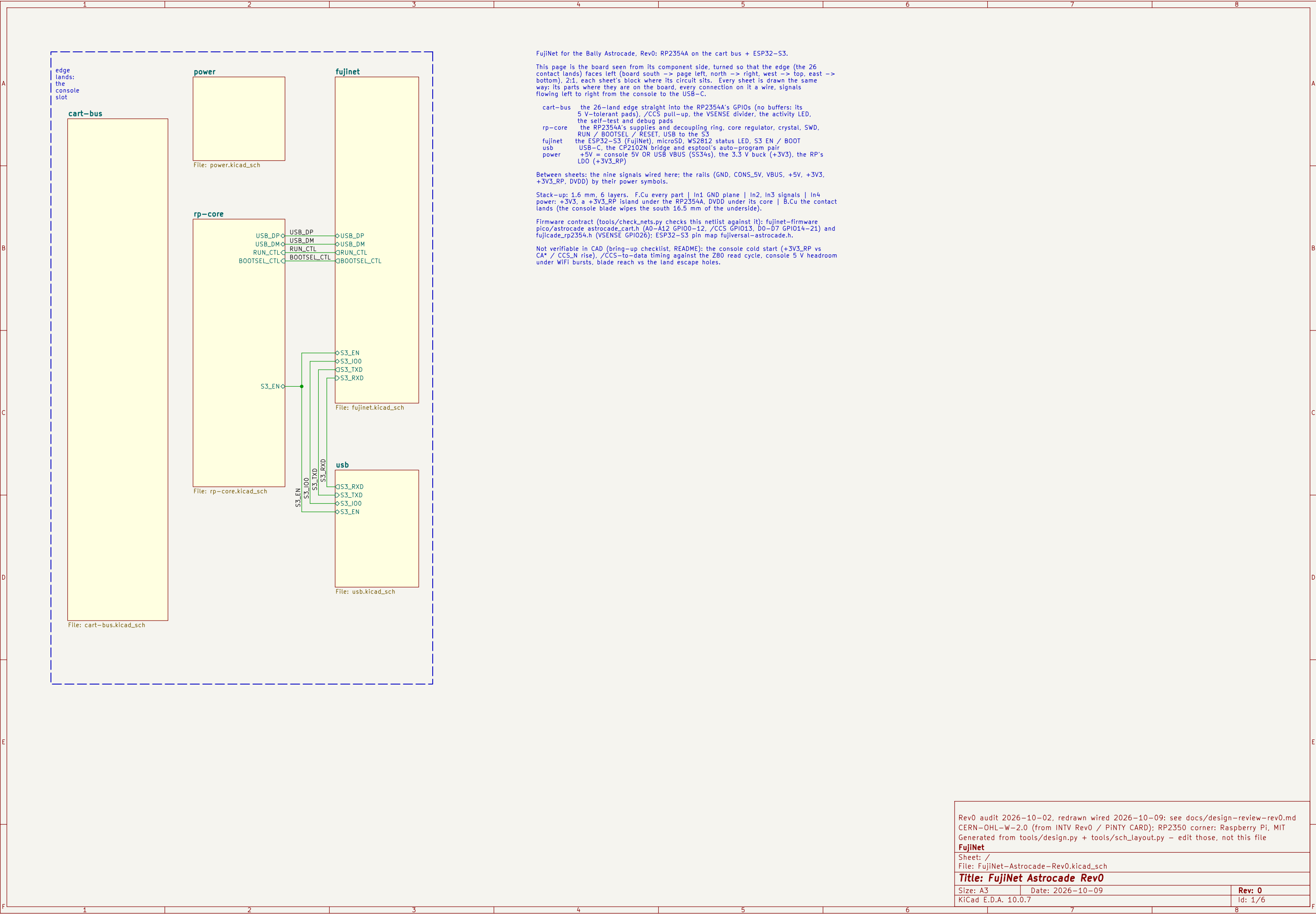
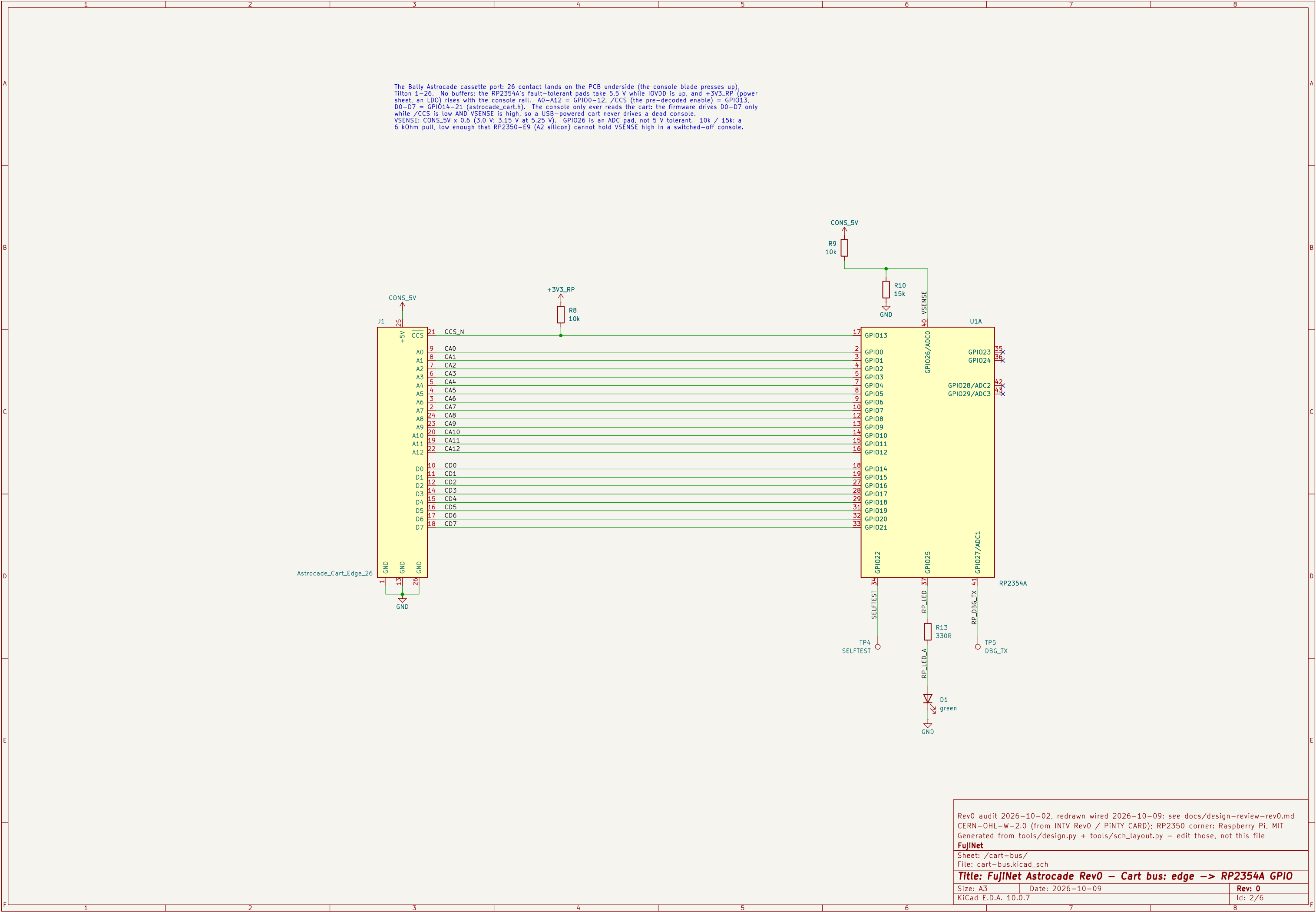
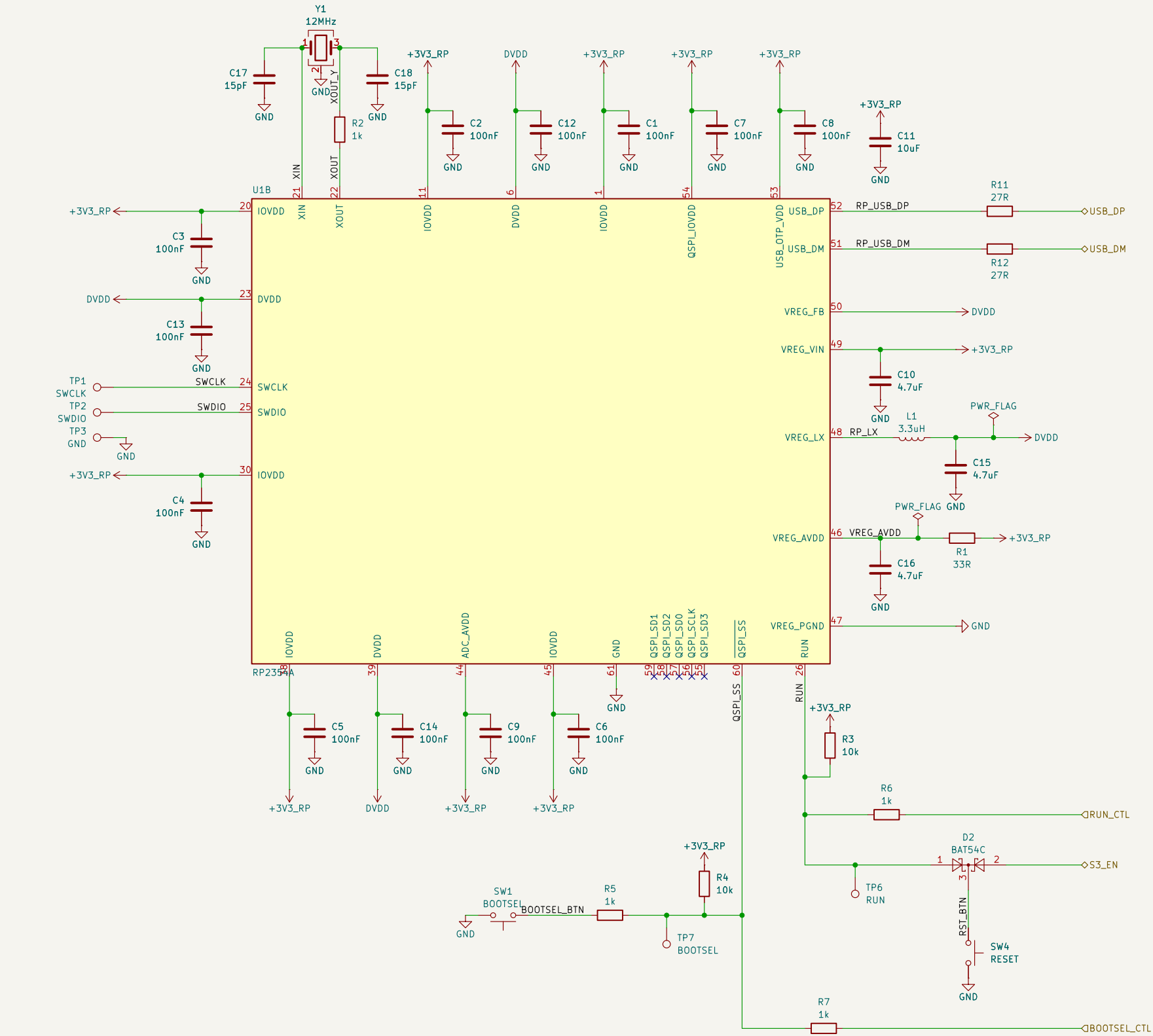


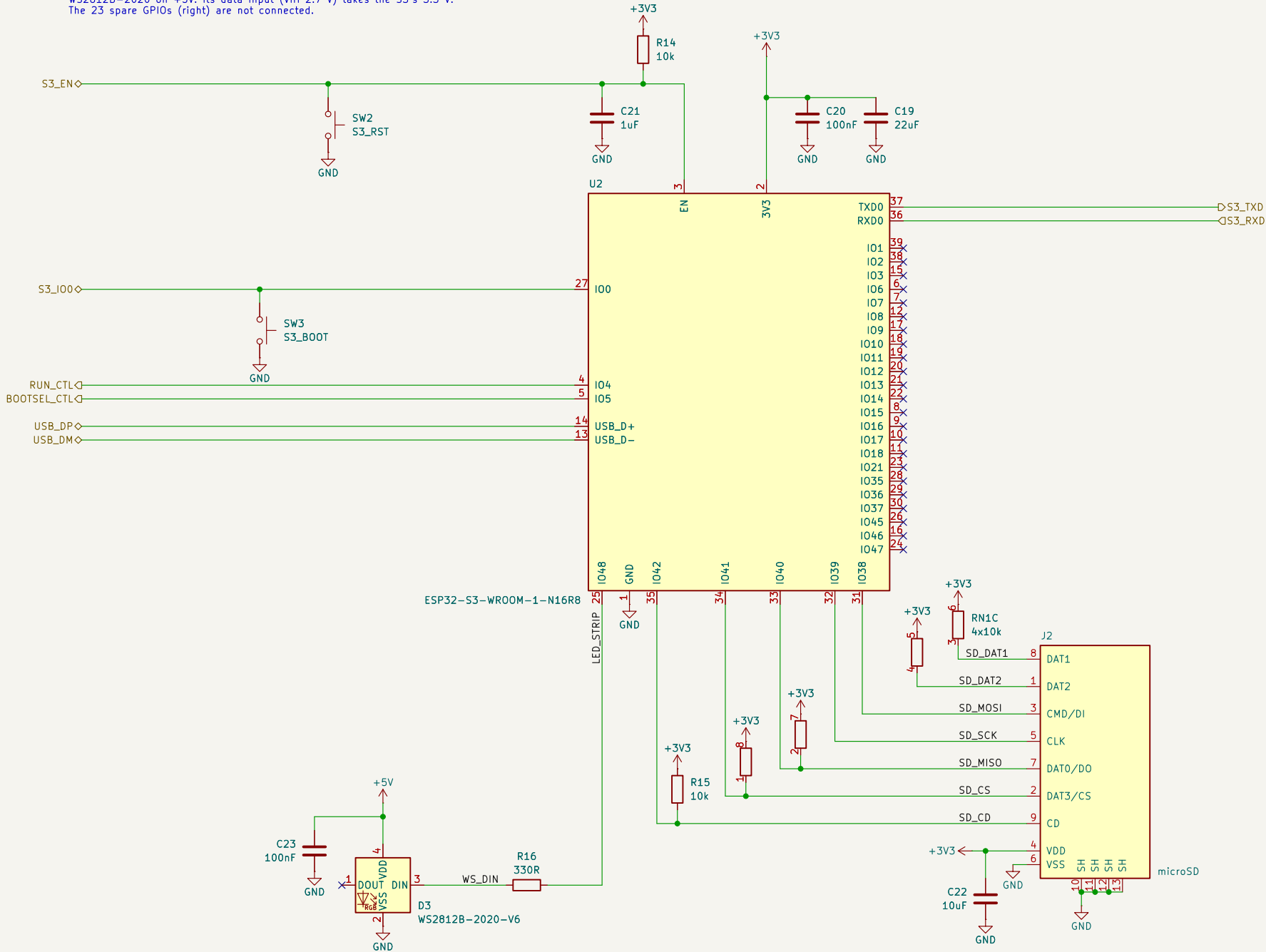


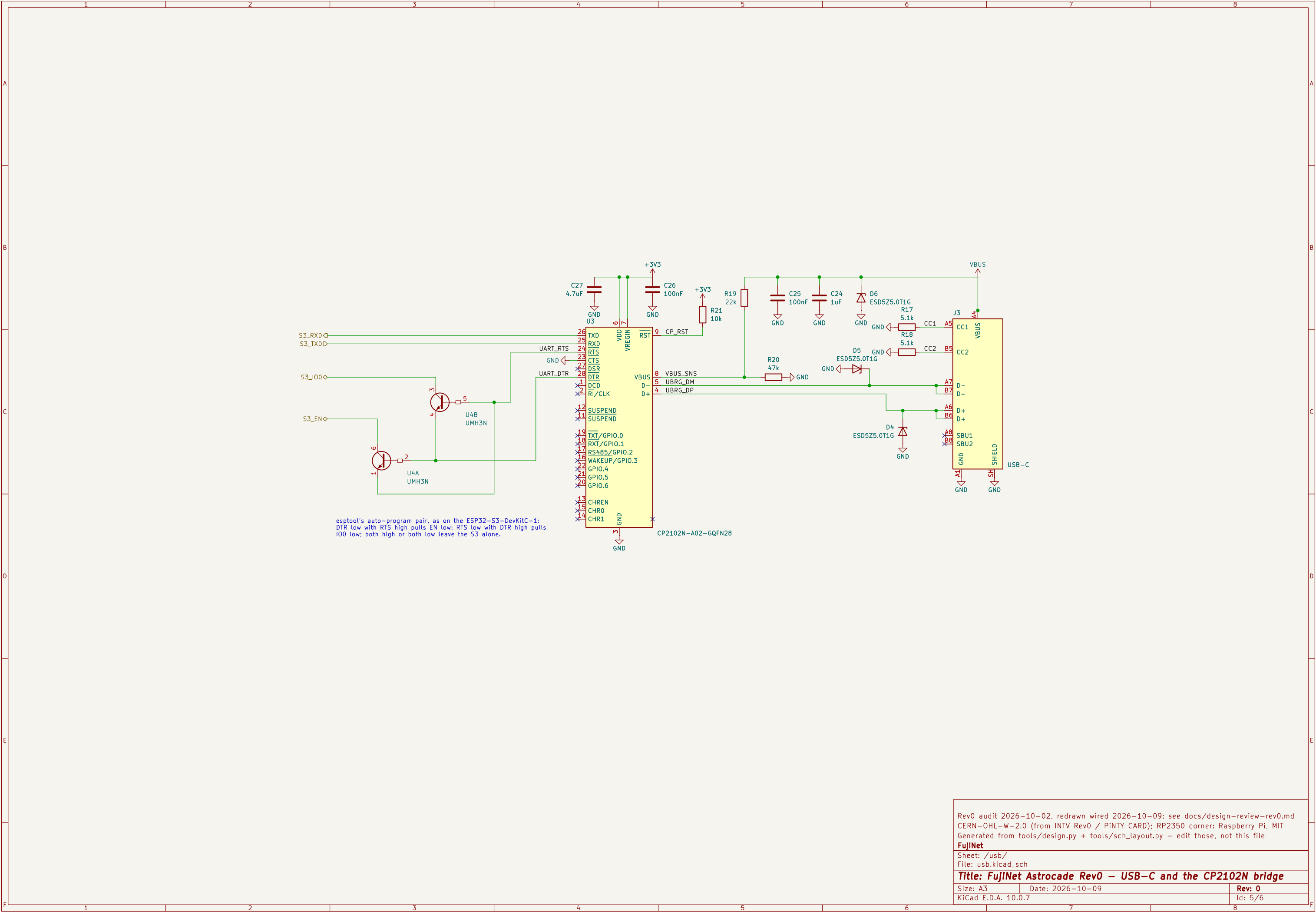


Supplies (RP2350 datasheet 6.3.7): IOVDD, QSPL_IOVDD, USB_OTP_VDD, ADC_AVDD and VREG_VIN on +3V3_RP, the AP2112K LDO that rises with +5V; the core regulator switches VREG_LX through 3.3 uH onto DVDD (VREG_FB senses it), its analog supply VREG_AVDD through 33R / 4.7 uF. One 100 nF at every supply pin, drawn at the pin it serves (on the board they reach it through the In4 +3V3_RP island and DVDD pour); the regulator corner is the Raspberry Pi RP2350A minimal design's. RESET resets the RP2354A and the ESP32-S3 together (D2 steers RST_BTN onto RUN and S3_EN). Hold BOOTSEL while pressing RESET for the RP's USB boot ROM. The S3 does both: IO4 (RUN_CTL) low resets the RP, IO5 (BOOTSEL_CTL) low through that reset selects BOOTSEL. QSPL_SD0-3 / SCLK unused; the 2 MB flash is in the package.

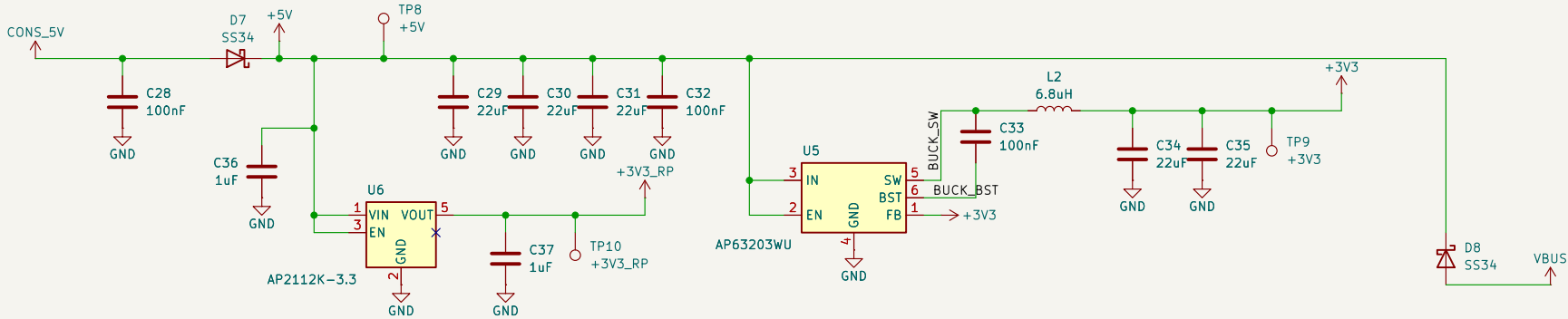


ESP32-S3-WROOM-1-N16R8: the FujiNet (fujiversal-astrocade.h pin map).
EN: 10k / 1 uF power-on delay. The S3_RST button, the cart RESET (rp-core, through the BAT54C) and the USB bridge's auto-program pair (usb sheet) all pull it low; S3_BOOT and the same pair pull I00. I04 / I05 drive the RP2354A's RUN and BOOTSEL; I019 / I020, the S3's own USB, go to the RP's. microSD in SPI mode, 10k pull-ups on CS, D0, DAT1, DAT2 and CD. WS2812B-2020 on +5V: its data input (VIH 2.7 V) takes the S3's 3.3 V. The 23 spare GPIOs (right) are not connected.

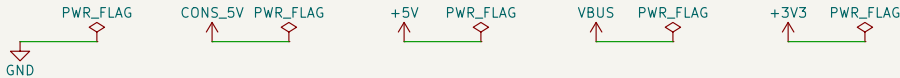




Power tree: +5V is the OR of two sources, so neither can back-feed the other.
CONS_5V edge land 25, the console's +5V, through D7 SS34.
VBUS USB-C, through D8 SS34.
SS34 VF -0.35 V at the cart's -0.4 A; +5V is 4.4-4.9 V; there is no 5 V logic on this board.
U6 AP2112K-3.3 LDO -> +3V3_RP: the WHOLE RP2354A (IOVDD, QSPI / USB / ADC supplies, VREG_VIN, VREG_AVDD).
In dropout it follows +5V as the console rail rises (1 V/ms into ~21 uF = 21 mA, under its 50 mA fold-back limit), so the 5 V-tolerant pads are never unpowered with the console bus driving them (RP2350 datasheet: the FT pads take 5.5 V 'provided IOVDD is powered to 3.3 V').
U5 AP63203 3.3 V / 2 A buck -> +3V3: ESP32-S3 (WiFi bursts ~400 mA), CP2102N, microSD; it waits for UVLO, then soft-starts over 4 ms. TP8, TP9 and TP10 are bring-up test points.



PWR_FLAG: these rails reach their loads only through passive parts (the edge land, USB-C, the SS34s, the buck's inductor).



Rev0 audit 2026-10-02, redrawn wired 2026-10-09: see docs/design-review-rev0.md
CERN-OHL-W-2.0 (from INTV Rev0 / PINTY CARD); RP2350 corner: Raspberry Pi, MIT
Generated from tools/design.py + tools/sch_layout.py - edit those, not this file

FujiNet

Sheet: /power/
File: power.kicad_sch

Title: FujiNet Astrocade Rev0 - Power: 5V OR, 3.3V buck, RP LDO

Size: A3	Date: 2026-10-09	Rev: 0
KiCad E.D.A. 10.0.7		Id: 6/6